

TED (10)4039
Revision 2010

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MODEL QUESTION
FIFTH SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL AND
ELECTRONICE ENGINEETING-----OCTOBER 2012.

INDUCTION MOTORS.

Time: 3hours.

(Maximum marks: 100)

PART—A

(Answer the following questions in one or two sentences. Each question carries 2 marks.)

5x2=10

- I
1. Why the air gap between stator and rotor of an induction motor is made very small?
 2. How can the direction of rotation of a 3 phase induction motor be reversed?
 3. What is the condition for maximum torque in running condition of an IM?
 4. What do you mean by circle diagram of an induction motor?
 5. Which type of motor is used in ceiling fan?

PART-B

II (Answer any five of the following. Each question carries 6 marks) 5x6=30

1. Explain the production of rotating magnetic field in a 3 phase induction motor.
2. Explain why the stator and rotor cores of an induction motor are made of magnetic Material..
3. Mention the effect of voltage of an induction motor operation.
4. Describe the speed control of induction motor by changing stator frequency.
5. Sketch and explain the working of an autotransformer starter
6. Explain the working principle of a single phase induction motor with the help of double field theory.
7. Draw and explain the working principle of repulsion start motor.

PART—C

(Answer one full question from each unit. Each question carries 15 marks) 4x15=60

MODULE—I

- III (a) Explain the construction and working principle of three phase Squirrel cage Induction motor. 8
- (b) A 3-phase, 6 pole, 50 Hz induction motor has a rotor resistance of 0.32 ohm per phase and standstill reactance of 0.8ohm per phase. The rotor voltage at stand still is 125 V per phase and rotor current is 16 A. Calculate the speed at which the motor is running.

OR

- IV (a) Prove that the frequency of the rotor induced in an induction motor is slip times its stator frequency 8
- (b) A 6 pole, 3-phase induction motor is connected across a 440V, 50Hz supply

source. Calculate the speed of the rotating magnetic field produced. What would be the speed of the rotor when slip is 0.04? Also calculate the frequency of rotor current at standstill and at a slip of 0.03.

7

MODULE—II

- V (a) Explain with the help of curves, the effect of variation of rotor circuit resistance on the torque-slip characteristic of an induction motor. 8
 (b) A particular 3-phase, 50 Hz, 4 pole, induction motor has rotor resistance of $0.04 \Omega/\text{phase}$. The maximum torque occurs at a speed of 1200 rpm. Calculate the starting torque as a percentage of maximum torque. 7

OR

- VI (a) Establish the relation, “rotor copper loss = Slip x Rotor input” for a polyphase induction motor. 8
 (b) A 4-pole, 50 Hz, 7.46 kW motor has, at rated voltage and frequency, a starting torque of 160 percent and a maximum torque of 200 percent of full load torque. Determine (1) full load speed (2) speed at maximum torque. 7

MODULE—III

- VII (a) Explain the construction and working of a double cage induction motor. 8
 (b).Explain why it is recommended that large induction motors be started with reduced voltage applied across its terminals at starting. 7

OR

- VIII (a) Draw the circle diagram for a 5.6 kW, 400-V, 3 phase, 4 pole, 50 Hz, Slip-ring induction motor from the following data:
 No-load reading: 400V, 6 A, $\cos \Phi_0 = 0.087$, 12
 Short circuit test: 100V, 12 A, 720 W
 The ratio of primary to secondary turns=2.62, stator resistance per phase is 0.67Ω and of the rotor is 0.185Ω . Calculate (1) full-load current (2) full-load slip.
 (3) full-load power factor (4) Maximum power
 (b) List the applications of induction motors. 3

MODULE-IV

- IX (a) Explain with diagram, the torque slip characteristics of single phase induction motor 8
 (b) Describe how a rotating magnetic field is accomplished in a shaded pole motor. 7
 OR
 X (a) Mention the advantages of a two-value capacitor type induction motor over a single value one. 8
 (b) Explain the nature of magnetic field produced when a single phase supply is applied across a single-phase stator winding. 7
